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INDUSTRIAL PRODUCT DATASHEET

reComputer Mini J4012 with Extension - Intelligent Edge AI Computer with NVIDIA® Jetson™ Orin™ NX 16GB

Hand-size edge AI computer with 100 TOPS performance for generative AI and video analytics powered by NVIDIA Jetson Orin NX 16GB.

SKU	UPDATED	PRICE
114993553	2026-07-30 05:41:01	$999.00(Excl. VAT)



Overview



The reComputer Mini J4012 with Extension is a compact edge AI computer featuring the NVIDIA Jetson Orin NX 16GB module, delivering up to 100 TOPS AI performance with low power and latency. It supports generative AI at the edge with Vision-Language Models and multi-stream video analytics, with pre-installed JetPack 6.0 and rich I/O for scalable deployment.

Key Features

Price Increase Announcement
Date: February 28, 2026

Due to surging demand for modules and SSDs, prices for reComputer and reServer Jetson devices will increase by approximately $50 to $400 in the coming weeks. Please confirm your requirements and place your orders promptly to secure your stock.

- **Brilliant AI Performance for Production:** Achieves up to **100 TOPS** AI performance with low power and latency, built by **NVIDIA Orin SoC** combining the NVIDIA Ampere™ GPU architecture with 64-bit operating capability, integrated advanced multi-function video and image processing, and NVIDIA Deep Learning Accelerators.
- **Hand-Size Edge AI Device:** Compact size at 63mm x 95mm x 66.7mm, featuring an NVIDIA **Jetson Orin NX 16GB** module, Mini J401 carrier board, fan and enclosure. Support desktop and wall mounting.
- **Expandable with Rich I/Os:** Includes up to 8x USB, DP 1.4, 1x RJ45 for GbE, M.2 Key E, M.2 Key M, dual channel CAN, and GPIO.
- **Accelerate Solutions to Market: Pre-installed JetPack 6.0** on 128GB NVMe SSD, Linux OS BSP, supporting Jetson software and leading AI frameworks.
- **Scale to deploy:** support OTA, remote management services powered by Allxon and Balena.
- **Flexible Customization:** inlcudes changing accessories modules, logo, and hardware interfaces modification based on reComputer Mini J4012 original design.

Product Gallery











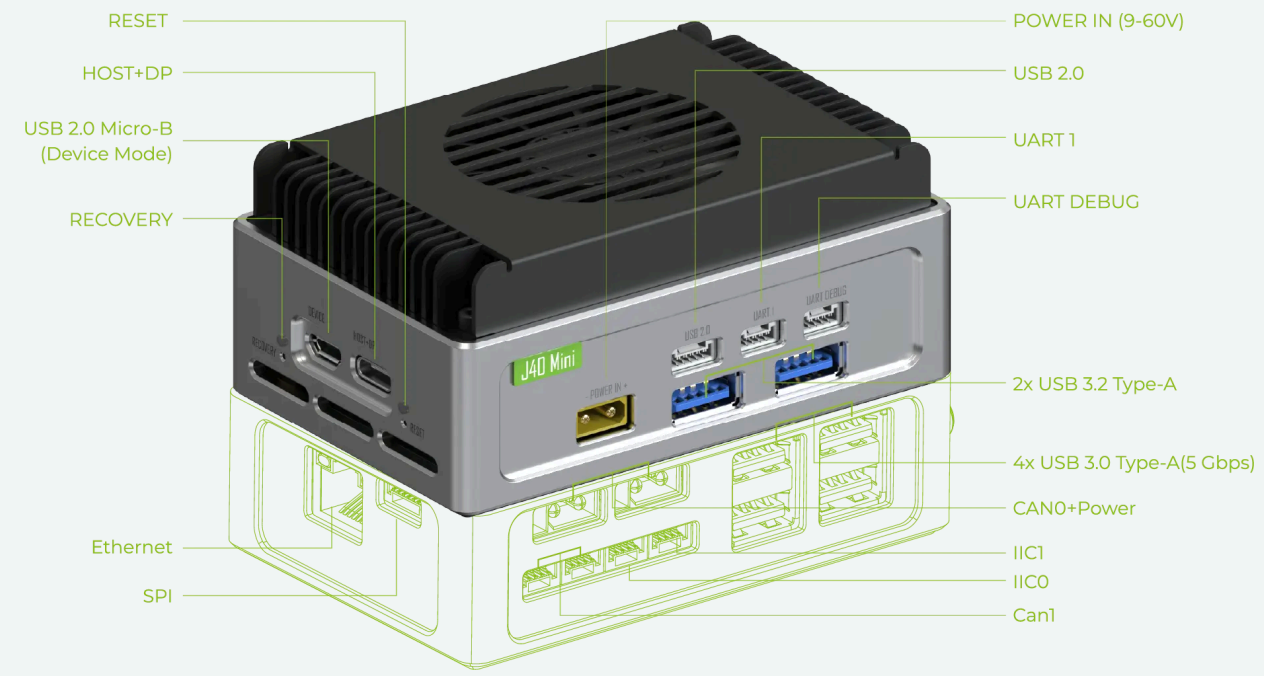


Specification

Jetson Orin System on Module

AI Performance	reComputer Mini J4012 with Extension - Orin NX 16GB, 100 TOPS		reComputer Mini J3011 with Extension - Orin Nano 8GB – 40 TOPS
GPU	1024-core NVIDIA Ampere architecture GPU with 32 Tensor Cores		
CPU	8-core Arm® Cortex®-A78AE v8.2 64-bit CPU (Orin NX 16GB)		6-core Arm® Cortex®-A78AE (Orin Nano 8GB)
Memory	16GB 128-bit LPDDR5 (Orin NX 16GB)		8GB 128-bit LPDDR5 (Orin NX 8GB)
Video Encode	1x 8K30 (H.265) | 2x 4K60 (H.265) | 4x 4K30 (H.265) | 9x 1080p60 (H.265) | 18x 1080p30 (H.265)		1080p30 supported by 1-2 CPU cores
Video Decode	1x 8K30 (H.265) 2x 4K60 (H.265) 4x 4K30 (H.265) 9x 1080p60 (H.265) 18x 1080p30 (H.265)		1x 4K60 (H.265) 2x 4K30 (H.265) 5x 1080p60 (H.265) 11x 1080p30 (H.265)
DL Accelerator	2x NVDLA v2		/
Carrier Board			
Storage	1x M.2 Key M PCIe (M.2 NVMe 2242 SSD 128G included)		
Networking	M.2 KEY E	1x M.2 Key E for WiFi/Bluetooth module	
I/O	USB	2x USB 3.2 Type-A (10Gbps); 1x USB 2.0 Micro-B (Device Mode); 1x USB 3.2 Type-C (Host Mode); 1x USB 2.0 JST-5pin (Host Mode);	
	UART	1x UART 1x Debug Uart	
	Display	1x DP 1.4 (included in Type-C conenctor)	
	Fan	1x 4 pin Fan Connector (5V PWM)	
	Button	1x Reset Button; 1x Recovery Button	
	Extension Port	2x 60 pin High Speed Connector (For extension board) 1x 10 pin Power Connector (For extension board)	
	RTC	1x RTC 2-pin	
Power	1 x XT30 Connector (12-54V DC)		
Extension Board			
Networking	Ethernet	1x RJ-45 Gigabit Ethernet (10/100/1000M)	
I/O	USB	4 x USB 3.0 Type-A (5 Gbps)	
	CAN (Providing Power)	CAN0:XT30(2+2)x2 CAN1:JSTx2	
	FAN	1x 4 pin Fan Connector (5V PWM)	
	I2C	2 x I2C JST 4pin (3.3V)	
	SPI	1 x SPI JST 6pin (3.3V)	
Mechanical	Dimensions (W x D x H)	63mm x 95mm x 42mm (Without Extension) 63mm x 95mm x 66.7mm (With Extension)	
	Weight	345g (Without Extension) 462g (With Extension)	
	Installation	Desk, Wall-mounting	
Operating Temperature	-10℃~50℃		
Warranty	1 Year		

Hardware Overview



Applications

- **Capable of Bringing Generative AI to the Edge:** Now you can build AI agents capable of processing large amounts of live or archived videos and images with Vision-Language Models (VLM) such as LLaVA, this new AI agents help nearly every industry summarize, search, and extract actionable insights from video using natural language.
- **Build Multi-streams AI Video Analytics:** For building video analytics, Jetson Orin NX incorporates the NVIDIA Multi-Standard Video Decoder. This video decoder accelerates video decode, supporting low resolution mobile content, Standard Definition (SD), High Definition (HD) and UltraHD (8K, 4K, etc.) reComputer J4012 can take 18x1080p30 streams. See our tested YOLOv8 performance using NVIDIA Deepstream of single model and multi model on multi streams.
- **Fastest Way to Deploy Generative AI and Computer Vision Models:** We have jetson-example prepared for you! It offers one-line deploy projects edge AI applications of generative AI including Ollama, Llama3; computer vision including YOLOv8; and others. We have configured all environment for you to provides single command deployment of projects.

Part List

reComputer Mini J4012 (with Extension)	x1
Antenna	x2
xt30 to DC power connector	x1

Compliance & Logistics



HSCODE	8471504090
USHSCODE	8517180050
UPC	
EUHSCODE	8471707000
COO	CHINA

FCC Requirement

Any changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:

(1) this device may not cause harmful interference, and

(2) this device must accept any interference received, including interference that may cause undesired operation.

15.21

This transmitter must not be co-located or operating in conjunction with any other antenna or transmitter.

Note: This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules.

These limits are designed to provide reasonable protection against harmful interference in a residential installation.

This equipment generates, uses, and can radiate radio frequency energy.

If not installed and used in accordance with the instructions, it may cause harmful interference to radio communications.

However, there is no guarantee that interference will not occur in a particular installation.

If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

Resource Links

- https://www.seeedstudio.com/reComputer-Mini-J4012-with-Extension-p-6353.html
- Datasheet
- Schematic
- 3D File
- Seeed Nvidia Jetson Product Catalog
- Nvidia Jetson Comparison
- Nvidia Jetson Product Comparison
- Seeed Nvidia Jetson Success Cases
- Seeed Jetson One Pager

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